

Flying in VR: A Therapist-Supervised Prototype for Aviophobia

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Abstract

We present a therapist-supervised VR flight-exposure station for studying aviophobia. The system combines a stand-alone Meta Quest Pro headset (with eye and face tracking) and a custom-built seat driven by three actuators via a Raspberry Pi controller. A web dashboard provides therapist monitoring and layered emergency stops, ensuring safe operation. The platform also enables synchronized data capture during natural scenario pauses. In a feasibility study, the prototype was rated highly usable, engaging, and well-tolerated, with minimal simulator sickness and no adverse events. These findings suggest that supervised, motion-enabled VR can serve as a safe foundation for future phobia research.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Empirical studies in HCI*.

Keywords

Virtual Reality; Exposure; Aviophobia; Motion Platform; Eye Tracking; Face Tracking; Emotion Recognition; HCI

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1 Introduction

Fear of flying, or aviophobia, affects millions of people worldwide and is often accompanied by anticipatory anxiety that can disrupt

both personal and professional life [Oakes and Bor 2010]. Traditional exposure therapy has shown effectiveness, yet arranging real flight sessions is prohibitively expensive, logistically complex, and not always feasible for patients or clinicians [Abuso et al. 2023]. Virtual Reality (VR) offers a promising alternative, providing immersive, controllable, and repeatable conditions that can simulate key aspects of air travel without leaving the ground [Shiban et al. 2017].

In this project, we set out to design and implement a complete, low-cost station to support research on VR-based flight exposure. The system integrates hardware, software, and data pipelines to (a) evoke core phases of flight through synchronized visual and physical cues, (b) log multimodal signals for later analysis, and (c) create a technical foundation for future clinician-supervised studies. This paper reports on the first integrated prototype and a feasibility pilot, with a focus on engineering transparency, calibration, and operational safety rather than therapeutic efficacy.

2 Background and Related Work

VR exposure therapy is an established and well-validated method for treating specific phobias, including fear of flying, with outcomes comparable to traditional in-vivo exposure while offering greater accessibility, safety, and patient acceptance [Maples-Keller et al. 2017]. Prior research has demonstrated that immersive simulations of flight phases—takeoff, turbulence, and landing—delivered entirely within VR can significantly reduce avoidance and anxiety symptoms [Rothbaum et al. 2006, 2000]. However, these systems typically rely solely on visual and auditory cues, without external augmentation. Our approach extends this line of work by integrating motion and haptic feedback to enrich the perceptual experience, aiming to increase realism, strengthen presence, and improve ecological validity for future therapeutic applications.

3 Technical Implementation

The station is built around a rigid seat frame equipped with aviation-style harnesses and a backrest-mounted vibration motor that provides engine-rumble and turbulence micro-cues. Global motion is generated by three electric linear actuators in a triangular configuration, enabling heave, pitch, and roll dynamics. Mechanical

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Figure 1: VR flight-exposure station. Left: virtual cabin environment as seen in the headset. Right: participant seated on the motion platform.

end-stops and limit switches support calibration and safety, while conservative software constraints limit actuator travel during user sessions. The 3-DOF seat uses electric linear actuators rated at 15 mm/s peak travel speed; the platform authority is approximately $\pm 10^\circ$ pitch, and ± 100 mm heave.

A Raspberry Pi controller manages actuators and relays, exposes REST endpoints over Wi-Fi for trajectory commands, telemetry, and emergency stop, and runs a watchdog process to prevent runaway motion. The motion system can be addressed directly from the HMD application as well as from a browser-based dashboard operated by the supervising researcher or therapist.

The VR application runs stand-alone on a Meta Quest Pro and casts the participant as a seated passenger aboard an aircraft. Exterior views are rendered to match each flight phase (horizon attitude, ground proximity, cloud layers, weather), while the motion platform supplies congruent vestibular cues. Ambient audio—engine hum, hydraulic actuations—and spatialized cockpit/crew announcements are time-aligned with visuals and motion to reinforce presence. Dynamic cabin-lighting cues (e.g., seatbelt sign, takeoff/landing dimming) synchronize with scenario state and turbulence levels to modulate expectancy and arousal. To maintain comfort and performance on-device, the scene employs foveated rendering, baked lighting, and level-of-detail (LOD) models, targeting 90 Hz. Although the virtual cabin is modeled on an MI-8 VIP transport helicopter, it was chosen for its ergonomic comfort and large window apertures that afford clear exterior views; flight-phase visuals and motion profiles were calibrated to fixed-wing dynamics (takeoff, climb, cruise/turbulence, approach, landing) to preserve ecological plausibility.

Scenarios cover the main phases of flight—takeoff, climb, cruise with mild or moderate turbulence, approach, and landing—with motion profiles that are phase-specific and amplitude-scaled. During takeoff and flare, micro-vibration and subtle camera jitter are applied in-phase with actuator motion to enhance the sense of runway texture and crosswind buffeting. Optional “anticipation” markers (e.g., illuminated seatbelt sign and chime) can precede turbulence segments to support graded exposure and therapist-controlled predictability.

A browser-based dashboard enables the supervising researcher or therapist to select scenarios, adjust intensity, pause or resume sessions, and trigger an emergency stop. Session metadata are automatically appended to logs.

4 Feasibility and Usability Study

We recruited 34 students from a technical university as flight-experience participants and 5 psychologists as mentors supervising sessions via the dashboard. Participants were screened for typical VR/motion contraindications and conditions that could increase risk (e.g., history of photosensitive epilepsy, severe motion sickness,

vestibular or recent musculoskeletal disorders). Only adults able to provide consent were admitted. Basic anonymized demographics (age range, gender self-report, prior VR/flight experience) were collected. After a safety briefing and informed consent, each participant completed one exposure session at conservative motion and audio-visual settings under mentor supervision.

The study prioritized feasibility and usability. From the participant perspective we assessed perceived realism, comfort, sense of safety, motion intensity, engagement, and overall satisfaction using brief Likert-scale items and a short semi-structured post-experience interview. From the mentor perspective we evaluated the web dashboard’s usability and perceived control, including task success for selecting scenarios, adjusting intensity, pausing/resuming, and invoking emergency stop; clarity of system state, trust and workload, and overall satisfaction, with interaction traces (e.g., control adjustments and timings) recorded to quantify efficiency and identify friction points. Standardized instruments included the System Usability Scale (SUS), NASA-TLX, the Trust in Automation scale [Jian et al. 2000], the Simulator Sickness Questionnaire (SSQ), the Igroup Presence Questionnaire (IPQ), and the User Engagement Scale–Short Form (UES–SF). In parallel, we stress-tested the real-time biosignal data path under field conditions—characterizing end-to-end throughput, achievable sampling rates, latency/jitter, and packet loss for synchronized sensor streams. These signals were used only to validate streaming capacity and synchronization at this stage. In future work the validated pipeline may support graded-exposure personalization or offline analyses (e.g., arousal detection). All procedures followed institutional ethical guidelines, data were pseudonymized and no directly identifying information was stored.

5 Results

In the combined sample of students and mentors ($n=39$), usability was high (SUS $M=86.2$, 95% CI [83.7, 88.6]; $89.7\% \geq 80$), and perceived workload remained low-moderate (NASA-TLX $M=28.8$, 95% CI [27.1, 30.5]). Reported Trust in the system was moderate-high (Trust in Automation $M=64.5$, 95% CI [62.9, 66.1]). Because mentors interacted only with the web dashboard (no headset exposure), the following were administered to students only ($n=34$): engagement (UES–SF overall $M=3.86$ on 1–5, $38.2\% \geq 4$), presence (IPQ overall $M=1.92$ on $-3 \dots +3$, $41.2\% \geq 2$), and cybersickness (SSQ Total $M=24.64$, IQR = 11.22, only $2.94\% \geq 40$). Note on ecological validity: while the MI-8 VIP cabin departs from a typical airliner interior, the choice prioritized visual affordances and seating comfort. Transferability to commercial airliner cabins will be verified in a fixed-wing cabin variant. For data acquisition, packetized logging buffered on the HMD and sent during natural scenario pauses (segments 1:52–3:56) maintained synchronized eye/face and system features at a steady 35 Hz on Meta Quest Pro without contention. Aggregated results and de-identified logs are available in an anonymized online repository for review.

6 Conclusions

Our therapist-supervised, motion-enabled VR flight-exposure station proved both feasible and usable, offering high perceived quality and engagement with manageable workload, strong user trust, and minimal adverse effects. The packetized data acquisition approach

worked reliably during sessions, maintaining 35 Hz synchronized logging. Looking ahead, we plan to extend testing to clinical cohorts, integrate therapist-in-the-loop titration, personalize motion and audio cues, and further validate the biosignal pipeline toward affective modeling.

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